

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE)

December 2012

CE301 Parallel Computing

Date: 27.11.2012, Tuesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Do not write content of Header.h file for any program.

SECTION-I

Q-1 Answer the following questions. [11]

[A] What are the return values of fork()? [2]

[B] What is difference between fork() and process_fork()? [2]

[C] How many total numbers of processes (Including Parent) will be created from following program? Justify your answer by taking suitable diagram. [2]

```
main()
{
    for(i=0;i<3;i++)
        for(j=0;j<=2;j++)
            fork();
}
```

[D] Why do we need high speed computing? [2]

[E] What are the disadvantages of Distributed Memory Architecture? [3]

Q-2 Answer the following questions. [12]

[A] Differentiate between data parallelism and temporal parallelism. [4]

[B] By taking suitable example, explain Java Remote method invocation. [4]

[C] Write a program which exhibits behavior of producer and consumer using fork and self-scheduling. [4]

OR

Q-2 Answer the following questions. [12]

[A] Write and explain barrier function call. [6]

[B] By taking suitable example, explain expression splitting. [6]

Q-3 Answer the following questions.(Attempt any Two) [12]

[A] Write a parallel program in 'C' to subtract two 4 X 4 matrix and store result into third matrix using loop-splitting and fork().

[B] Write a parallel program in 'C' to count number of characters, words of a given string using self-scheduling and fork().

[C] Write a parallel program in 'C' to create histogram from an array using fork() & self-scheduling.

SECTION-II

- Q-4 Answer the following questions.** [11]
[A] What are the attributes of thread? [2]
[B] What are the disadvantages of RPC? [2]
[C] What are the ways to locate address of server in DCE environment? [2]
[D] Write and explain arguments of spawning of children. [2]
[E] What are the three types of operating system for multiprocessors? What are the characteristics of each? [3]

- Q-5 Answer the following questions.** [12]
[A] What are the differences between Cluster, Grid and Cloud Computing? [4]
[B] Write a thread program to count number of vowels in a given text. [4]
[C] What is condition variable in thread? Write the advantages of it. [4]

OR

- Q-5 Answer the following questions.** [12]
[A] What is DCE? What is DCE Cell? Draw architecture of DCE. Explain services offered by DCE. [6]
[B] Explain the architecture of PVM and explain terminology used in PVM. [6]

- Q-6 Answer the following questions.(Attempt any Two)** [12]
[A] Write a thread program to perform matrix multiplication (3X3) using self-scheduling.
[B] Write down the steps to create Server service in DCE application. Explain each step in brief.
[C] Write a thread program to create partial histogram from an array using loop-splitting.